

METRO MOBILITY INNOVATION GROUP
Engineering & Operations Division

VoltBus Autonomous Electric Transit Initiative
Master Operations & Engineering Brief

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Executive Summary & Program Objectives

Program Overview

VoltBus is MMIG's flagship autonomous electric bus program, built to prove that a fully electric, self-driving transit fleet can run safely and reliably on real city streets.

The program is led by Chief Systems Architect Elena Rostova, whose team owns final sign-off on every safety and engineering decision before a change reaches passenger service.

- Electric propulsion — zero tailpipe emissions on every route
- Autonomous driving — supported by redundant sensors and a dedicated safety system
- Smart stations — real-time passenger info and charging integration

Program Objectives

- Prove reliable, all-electric transit service on a busy urban corridor
- Validate the onboard safety system under real traffic and pedestrian conditions
- Build a repeatable depot maintenance model ready for fleet-wide scale-up
- Collect the safety data needed for full autonomy certification

Key System Milestones

Phase	Status	What It Covers
Phase I — Platform Validation	Complete	Chassis and battery certification, closed-course autonomy testing, depot setup
Phase II — Controlled Pilot	Active	Live service on Route 101 with a safety operator onboard at all times
Phase III — Network Expansion	Planned	Rollout to five more corridors and removal of the onboard safety operator

Where We Are Today

The program is currently in Phase II. To move into Phase III, VoltBus needs 90 straight days of stable performance on Route 101 with zero unresolved safety incidents.

Vehicle Architecture & Power Systems

Chassis Overview

Each VoltBus is a 12-meter, low-floor urban bus, purpose-built for electric power and autonomous operation.

Spec	Detail
Length	12.0 meters
Passenger Capacity	84 (33 seated, 51 standing)
Drive System	Dual rear electric motors, 160 kW each

Battery & Charging

- 420 kWh total capacity, split across four LFP-2 (lithium-iron-phosphate) battery packs mounted under the floor
- LFP-2 chemistry was chosen for its strong thermal stability under repeated fast charging
- Charging happens through a top-mounted pantograph that connects to overhead charging heads at depots and stations
- 450 kW rapid charging restores about 60% battery in roughly 12 minutes

Thermal Management — Keeping Batteries Safe

Battery temperature is tracked continuously. Three simple rules govern how the system responds:

State	Temperature	What Happens
Normal	15°C – 40°C	No restrictions — normal charging and driving
Level 1 Warning	48°C	Cooling pumps jump to 100%; charging speed is reduced
Level 2 Emergency	58°C	Protocol Thermal-Shield activates: affected battery isolated, power capped at 30%, bus routed to Depot-Gamma

Why This Matters

Once Protocol Thermal-Shield triggers, it cannot be overridden by the driver or dispatch. Only Depot-Gamma technicians can clear a bus to return to service after a Level 2 event. This keeps the fleet safe even under schedule pressure.

Sensor Fusion & Autonomous Navigation

Sensor Suite

Every VoltBus sees its surroundings through three types of sensors working together:

Sensor	Count	What It Does
LiDAR (128-beam)	4	Builds a 3D map of the road and everything on it
HDR Cameras	8	Reads lanes, traffic signals, and identifies pedestrians
Radar	2	Tracks objects in rain, fog, or low visibility

Onboard Compute

- An Edge AI Compute Node runs Perception-OS v4.2, combining all sensor data into one live picture of the road, updated 20 times per second
- A backup compute module stands ready at all times — if the primary fails, the backup takes over in just 40 milliseconds

Safety Override Engine (SOE)

The SOE is a separate safety system that watches for pedestrians and obstacles independently of the main driving software. This separation means a glitch in the main system can never disable emergency braking.

- At 40 km/h, the SOE watches an 8-meter braking zone directly ahead of the bus
- If something enters that zone, the SOE instantly triggers both regenerative and pneumatic braking together
- **Result:** the bus comes to a full stop in just 2.1 seconds
- The driver can always brake manually at any time, on top of the SOE

Built-In Accountability

Every time the SOE activates, the event is logged and automatically sent to the MMIG Safety Board for review within one business day.

Route 101 Pilot Data & Incident Analysis

Route 101 Overview

- 14.2 km corridor connecting the northern districts to downtown
- 12 smart stations with real-time arrival info and, at 4 stops, charging heads
- Dispatched from the Metro Terminal Hub, led by Operations Supervisor Marcus Vance

Incident Case Study — July 12 Pilot Run

During a regional heatwave, VoltBus Unit #09 triggered a Level 1 Thermal Warning while stopped at Stop 7 (Oak Street).

Detail	Value
Vehicle	VoltBus Unit #09
Location	Stop 7 — Oak Street
Conditions	39°C, direct sun, heatwave
Cause	Two back-to-back 450 kW rapid charges in 45 minutes
Alert Level	Level 1 — Warning

How It Was Resolved

- Cooling pumps automatically ramped to 100%, as designed
- Charging speed was reduced until the battery cooled down
- **Unit #09 stayed in service** throughout — no passengers were affected and the route stayed on schedule
- **Temperature normalized in 14 minutes**, well before it could escalate to a Level 2 Emergency

Follow-Up Action

Engineering, under Elena Rostova, issued a new rule: during heatwaves above 35°C, Stop 7 allows only one rapid charge per layover instead of two. Since this change, no further thermal warnings have occurred on Route 101.

Infrastructure Expansion & Maintenance Protocols

Facility Roles

Facility	District	Handles
Depot-Alpha	North	Chassis, mechanical assemblies, structural checks
Depot-Gamma	South	Battery conditioning, sensor calibration, software updates

Any bus that trips a Level 2 thermal event is automatically routed to Depot-Gamma for inspection before it can return to service.

Maintenance SOP — Battery Cell Imbalance

Lead Technician David Miller maintains the standard procedure for handling battery cell imbalances at Depot-Gamma. If a battery pack shows more than 5% internal resistance imbalance between cell groups, it's automatically flagged for inspection.

- **Step 1 — Isolate:** the affected battery pack is disconnected and put into diagnostic mode
- **Step 2 — Run DR-88:** Diagnostic Routine 88 tests each cell group's resistance and capacity against its original baseline
- **Step 3 — Classify:** the pack is marked as fine, degraded (replace within 30 days), or faulty (replace immediately)
- **Step 4 — Sign off:** results are logged and countersigned by Technician Miller before the bus returns to Marcus Vance's dispatch pool

No Shortcuts

A bus can never return to service after a DR-88 check without a documented pass — no exceptions, even under schedule pressure.

What's Next

- Depot-Beta (East District): new facility in progress ahead of Phase III
- Two additional 450 kW charging heads planned for the Metro Terminal Hub
- Ongoing grid capacity review with the Regional Utilities Commission